

Deep Learning

Python Environments for Deep Learning

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Mandatory Packages (Course Baseline)

Required libraries

- TensorFlow
 - CPU: `tensorflow`
 - GPU: `tensorflow` with CUDA support
(requires NVIDIA GPU + drivers)
- `numpy`, `scipy`, `scikit-learn`, `matplotlib`
- `jupyter`

Rule

Always **activate the environment first**, then install packages.

requirements.txt (MANDATORY)

Create a file named `requirements.txt` in your project folder

```
tensorflow==2.20.0  
numpy==1.26.4  
scipy==1.14.1  
scikit-learn==1.8.0  
matplotlib==3.10.8  
jupyter==1.0.0
```

Do not change versions

These versions are pinned to avoid dependency conflicts.

Windows: Create and activate a venv

Step 1 — Create the environment (Python 3.11)

```
py -3.11 -m venv dl-venv
```

Step 2 — Activate (PowerShell)

```
.\dl-venv\Scripts\activate
```

Visual check

Your prompt should start with **(dl-venv)**.

Windows: Install mandatory packages (CPU)

Step 3 — Install packages (with env activated)

```
python -m pip install --upgrade pip
```

```
pip install -r requirements.txt
```

Common mistake

If you forget activation, packages install to the wrong Python.

Windows: Quick verification

Step 4 — Verify imports

```
python -c "import tensorflow as tf; print('TF', tf.__version__)"  
python -c "import numpy, scipy, sklearn, matplotlib; print('OK')"
```

Conda: Create and activate environment

Step 1 — Create (Python 3.11)

```
conda create -n dlcourse python=3.11 -y
```

Step 2 — Activate

```
conda activate dlcourse
```

Note

In this course, we use `pip` to install TensorFlow and the required Python packages.

Conda: Install mandatory packages (CPU)

Step 3 — Install packages (CPU baseline)

```
python -m pip install --upgrade pip
```

```
pip install -r requirements.txt
```

Conda: Verification

Step 4 — Verify imports

```
python -c "import tensorflow as tf; print('TF', tf.__version__)"  
python -c "import numpy, scipy, sklearn, matplotlib; print('OK')"
```

macOS: Requirements

Supported systems

- macOS laptop (Intel or Apple Silicon)
- Python 3.11 installed (via python.org or Homebrew)
- No native NVIDIA GPU support

Important

TensorFlow GPU is **not supported** on macOS for this course. Use CPU installation only.

macOS: Install Python 3.11 (if needed)

Option 1: python.org installer

Download from: <https://www.python.org/downloads/>

Option 2: Homebrew

```
brew install python@3.11
```

macOS: Create and activate virtual environment

Step 1 — Create

```
python3.11 -m venv dl-venv
```

Step 2 — Activate

```
source dl-venv/bin/activate
```

Visual check

Your prompt should begin with **(dl-venv)**.

macOS: Install mandatory packages (CPU)

Install packages

```
python -m pip install --upgrade pip
```

```
pip install -r requirements.txt
```

macOS: Verification

Verify installation

```
python -c "import tensorflow as tf; print('TF', tf.__version__)"  
python -c "import numpy, scipy, sklearn, matplotlib; print('OK')"
```

Ubuntu: Install Python 3.11 (if needed)

If `python3.11-venv` is missing

Add the deadsnakes repository first.

Install Python 3.11

```
sudo apt update
sudo apt install -y software-properties-common
sudo add-apt-repository -y ppa:deadsnakes/ppa
sudo apt update

sudo apt install -y python3.11 python3.11-venv python3.11-dev
```

Ubuntu: Create environment and install (CPU)

Step 1 — Create + activate

```
python3.11 -m venv dl-venv  
source dl-venv/bin/activate
```

Step 2 — Install mandatory packages

```
python -m pip install --upgrade pip  
  
pip install tensorflow  
pip install numpy scipy scikit-learn matplotlib  
pip install jupyter
```

Re-install the environment (clean reset)

When to re-install

Use a clean reset if installs are broken, imports fail, or versions got mixed.

Windows (PowerShell)

```
deactivate  
rmdir /S /Q dl-venv  
py -3.11 -m venv dl-venv  
.\dl-venv\Scripts\activate  
python -m pip install --upgrade pip  
pip install -r requirements.txt
```

Re-install (Ubuntu)

Ubuntu

```
deactivate  
rm -rf dl-venv  
python3.11 -m venv dl-venv  
source dl-venv/bin/activate  
python -m pip install --upgrade pip  
pip install -r requirements.txt
```

Re-install (Conda)

Conda

```
conda deactivate  
conda env remove -n dlcourse  
  
conda create -n dlcourse python=3.11 -y  
conda activate dlcourse  
python -m pip install --upgrade pip  
pip install -r requirements.txt
```

GPU Hardware Requirements

Minimum requirements

- NVIDIA GPU (CUDA-capable)
- Recommended: 8 GB VRAM or more
- Recommended: 16 GB system RAM
- Free disk space: 10–20 GB

If you do not have an NVIDIA GPU

Use the CPU installation.

Ubuntu (native): GPU installation (venv)

Step 1 — Check GPU visibility

```
nvidia-smi
```

Step 2 — Install TensorFlow GPU support

```
pip install "tensorflow[and-cuda]"
```

Step 3 — Verify

```
python -c "import tensorflow as tf;  
          print(tf.config.list_physical_devices('GPU'))"
```

Windows: GPU installation (recommended via WSL2)

Recommended

Use WSL2 + Ubuntu for modern TensorFlow GPU on Windows.

Inside WSL2 Ubuntu

```
nvidia-smi  
pip install "tensorflow[and-cuda]"  
python -c "import tensorflow as tf;  
          print(tf.config.list_physical_devices('GPU'))"
```

Conda + GPU (recommended in Linux/WSL2)

After activating your conda env

```
conda activate dlcourse
nvidia-smi
pip install "tensorflow[and-cuda]"
python -c "import tensorflow as tf;
           print(tf.config.list_physical_devices('GPU'))"
```